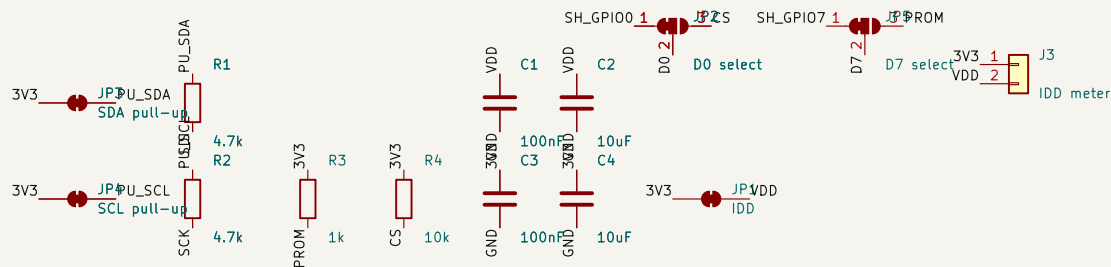


Board B: XIAO ESP32-S3 carrier for a Bosch Shuttle Board 3.0. Shuttle on top, XIAO underneath. 3.3 V only.
BME690 8x shuttle: D0–D7 (GPIO1, 2, 3, 4, 5, 6, 43, 44) are the chip selects of sensors U1–U8;
D8 SCK (GPIO7), D9 SDO/MISO (GPIO8), D10 SDI/MOSI (GPIO9). The XIAO Sense's microSD card shares
D8–D10 (its own CS is GPIO21), so on the 8x board the card and the sensors take turns on one SPI bus.

P1		P2	
VDD	1	VDD	1
3V3	2	CS	11
GND	3	SCK	12
SH_GPIO0	4	SDO	13
GPIO1	5	SDI	14
GPIO2	6	GPIO4	15
GPIO3	7	GPIO5	16
		GPIO6	17
		SH_GPIO7	18
		PROM	19
		PROM_RW	20
		Shuttle	3.0 sockets

D0	1	D0/GPIO1	U1	5V	14
GPIO1	2	D1/GPIO2	GND	13	GND
GPIO2	3	D2/GPIO3	3V3	12	3V3
GPIO3	4	D3/GPIO4	GPIO9	11	SDI
GPIO4	5	D4/GPIO5	GPIO8	10	SDO
GPIO5	6	D5/GPIO6	GPIO7	9	SCK
GPIO6	7	D6/GPIO7	GPIO44	8	D7
		D7/GPIO8	XIAO ESP32S3		



Selector jumpers, set for the 8x shuttle as made (pads 1–2 joined):
JP2: D0 → shuttle GPIO0 (8x: sensor U1 CS). Cut 1–2, bridge 2–3; D0 → shuttle CS (single-sensor shuttles).
JP5: D7 → shuttle GPIO7 (8x: sensor U8 CS). Cut 1–2, bridge 2–3; D7 → PROM_RW (shuttle ID chip).
JP1 IDD: cut and put a meter across J3 to measure the sensor VDD current.
JP3 / JP4: 4.7k I2C pull-ups on SDA / SCL: cut to remove.

R3 1k: PROM_RW pull-up for the DS28E05 1-Wire ID chip (datasheet: 300–1500 ohm at 3.3 V, overdrive speed only). R4 10k: CS pull-up so single-sensor shuttles start in I2C mode. VDDIO is tied to the XIAO's 3V3 (its logic level), so there is no TIE jumper on this board.

github.com/leakydata/bme690-toolkit
Sseed XIAO ESP32-S3 / Sense carrier for a Bosch Shuttle Board 3.0

Sheet: /

File: board_b.kicad_sch

Title: Board B – XIAO ESP32-S3 shuttle carrier

Size: A4 Date: 2026–10–03

KiCad E.D.A. 10.0.6

Rev: 1

Id: 1/1